

Apex Systems

Precision engineering. Proven security.

Proposal in response to

RFP-2026-041 — Cloud-based Real-Time Passenger Information & Fleet Operations Platform

Metro Transit Authority (MTA)

Submission date: September 10, 2026

1. Executive Summary

Apex Systems is pleased to respond to RFP-2026-041. With 12 years of experience delivering mission-critical transit technology across 40+ public-sector projects, we propose Apex TransitCloud, our proven real-time passenger information platform, tailored to the Metro Transit Authority's 500-vehicle fleet. Our solution is engineered for precision: a cloud-native microservices architecture with true real-time vehicle tracking, an accessible passenger mobile app, 120 digital station signs, and a dispatch operations dashboard — all backed by enterprise-grade security.

We offer a fixed price of \$485,000 for full delivery and year-one support, with a 9-month implementation timeline from kickoff to final go-live. We acknowledge this is premium pricing relative to the market; it reflects our higher price point for certified security controls, redundant infrastructure, and senior engineering staff assigned full-time to your program.

2. Understanding of Requirements

MTA requires a single platform that unifies real-time bus location data, disseminates it to riders via mobile and station signage, and gives dispatchers a live operational picture. Key requirements include GTFS-RT open standards compliance, integration with the existing CAD/AVL and fare collection systems, 99.9% platform uptime, WCAG 2.1 AA accessibility, and full go-live within 12 months. Our proposal addresses each requirement explicitly in the sections below.

3. Proposed Solution & Technical Approach

Apex TransitCloud is built on a cloud-native microservices architecture orchestrated with Kubernetes across multiple availability zones for full redundancy. Vehicle telemetry flows through an event-driven streaming pipeline (Apache Kafka) achieving sub-second real-time updates from bus to rider screen. The platform is horizontally scalable from 500 to over 2,000 vehicles without architectural changes.

Integration is a core strength. We expose a versioned REST API plus webhooks, and our integration layer ships with pre-built connectors for major CAD/AVL suites; a dedicated integration workstream will map MTA's legacy interfaces during discovery. All data exchanges follow open standards (GTFS-RT, SIRI) so the Authority avoids vendor lock-in. Station signs run a resilient edge client that caches

content locally and continues operating through network outages.

- Microservices architecture on Kubernetes; independently deployable, independently scalable services.
- Event-driven real-time pipeline: median end-to-end latency under 2 seconds.
- Open API with versioned REST endpoints, webhooks, and GTFS-RT/SIRI feeds.
- Redundant multi-AZ deployment with automated failover and daily encrypted backups.

4. Implementation Plan

Our 9-month timeline is organized into four phases with clear milestones, a full-time senior staffing plan, and a managed pilot before fleet-wide go-live. A named project manager and solution architect are assigned 100% for the program duration.

Table 1 — Implementation milestones

Phase	Duration	Key milestones / deliverables
1. Discovery & Design	6 weeks	Requirements discovery sessions; CAD/AVL integration mapping; architecture sign-off
2. Build & Integrate	16 weeks	Platform configuration; API integration; mobile app + signage build; milestone: system demo
3. Pilot	6 weeks	50-bus pilot; dispatcher training; milestone: pilot acceptance
4. Rollout & Go-live	8 weeks	Fleet-wide deployment; hypercare; milestone: final acceptance & go-live

Staffing: 1 project manager, 1 solution architect, 4 senior engineers, 1 QA engineer, 1 UX specialist. Risks are managed through a joint risk register reviewed bi-weekly; our top mitigated risks are CAD/AVL interface delays (buffered in Phase 2) and signage hardware lead times (ordered at kickoff).

5. Price Proposal

We propose a fixed price of \$485,000. Our cost breakdown is fully transparent, and our assumptions are documented below so the Authority can evaluate total cost with confidence.

Table 2 — Price breakdown (fixed price)

Item	Amount (USD)
TransitCloud platform licenses (3 years)	\$180,000
Implementation services	\$210,000
Signage hardware integration (120 signs)	\$45,000

Item	Amount (USD)
Training & documentation	\$20,000
Year-1 support & maintenance	\$30,000
Total fixed price	\$485,000

- Assumption: MTA provides CAD/AVL API access and test environment by week 3.
- Assumption: signage mounting and power are provided by MTA facilities.
- Assumption: up to 40 hours of stakeholder working sessions; excess billed at \$195/hour.
- Payment terms: 20% kickoff, 40% at pilot acceptance, 40% at final go-live.

6. Security, Compliance & Risk

Security is independently certified, not merely asserted. Apex Systems is ISO 27001 certified and completes a SOC 2 Type II audit annually; reports are available under NDA. All data is protected with AES-256 encryption at rest and TLS 1.3 in transit. We conduct annual third-party penetration tests and maintain a 30-day remediation SLA for critical findings.

Access control follows least-privilege role-based access control with SSO/MFA integration to the Authority’s identity provider. Our data privacy program aligns with GDPR and CCPA principles: rider data is minimized, anonymized for analytics, and never sold. Immutable audit logging supports full auditability of dispatcher actions and data exports.

7. Support Model, Experience & References

Apex provides a dedicated account manager, an 8x5 helpdesk with 4-hour response SLA, and 24/7 on-call coverage for severity-1 incidents. Our support model includes quarterly service reviews and proactive platform health monitoring.

Relevant experience includes three similar projects: real-time fleet tracking for Harbor City Transit (600 vehicles, 2023), a passenger information rollout for Northline Rail (2024), and a dispatch modernization program for Capital Metro (2025). References available on request.

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